Compiled Solutions – Senior Division

1. **00-01 C3 Digital Electronics**

$\overline{(\overline{A}B+\overline{B})\oplus\overline{B+\overline{C}}}$

(0,0,1) (1,0,1) (1,1,0) (1,1,1)

A	B	C	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{A}B$	$\overline{A}B+\overline{B}$	$B+\overline{C}$	$\overline{B+\overline{C}}$	$\oplus$ (both the same)
0	0	0	1	1	1	0	1	1	0	0
0	0	1	1	1	0	0	1	0	1	1
0	1	0	1	0	1	1	1	1	0	0
0	1	1	1	0	0	1	1	1	0	0
1	0	0	0	1	1	0	1	1	0	0
1	0	1	0	1	0	0	1	0	1	1
1	1	0	0	0	1	0	0	1	0	1
1	1	1	0	0	0	0	0	1	0	1

2. **00-01 C3 Digital Electronics**

The translation is as follows: $\overline{(A\oplus B)(\overline{B\oplus C})}$

3. **00-01 C4 Digital Electronics**

The circuit translates as follows: $(A+AB+(\overline{B}+C))(\overline{B}+C)(\overline{C+D})$ 2

To be true all 3 factors must be TRUE.

$\overline{C+D}$ is only TRUE for the quadruple (*, *, 0, 0),

but when B = 1, the second factor is FALSE.

Therefore, the choices are limited to (*, 0, 0, 0).

4. **01-02 C3 Digital Electronics**

The circuit translates as follows: $\overline{A\overline{B}+C}$. This simplifies to: (*,1,0) and (0,0,0)

$\overline{A}\overline{\overline{B}}\overline{C}=(\overline{A}+B)\overline{C}$. To be TRUE, both factors must be TRUE. OR

This implies that C must equal 0. The first factor gives 2 choices. (0,1,0), (1,1,0),

If A = 0, then B can be either 1 or 0. If A = 1, then B must equal 1. and (0, 0, 0).

5. **01-02 C3 Digital Electronics**

The circuit translates to: $(\overline{A+B})((B\oplus C)D)$. If B is FALSE, $\overline{A}CD$

$(\overline{A+B})=\overline{A}$, $B\oplus C=C$ so the entire expression is $\overline{A}CD$.

6. **01-02 C4 Digital Electronics**

The circuit translates: $\overline{(AB+\overline{C})(\overline{AB})}$ (0,0,1), (0,1,1), (1,0,1), (1,1,1), (1,1,0)

To be true one or both of the terms must be true. This occurs for (*, *, 1) and (1, 1, 0).

7. **02-03 C3 Digital Electronics**

The circuit translates as follows: $\overline{(AB\oplus(B+\overline{C}))C}$ (0, 1, 0)

$\overline{(AB\oplus(B+\overline{C}))C}=0$ implies $(AB\oplus(B+\overline{C}))C=1$ $\overline{C}=1$ so $C=0$. The XOR expression must = 1 also. If B = 0 both operands are FALSE and the expression is FALSE which leads to a contradiction. If B = 1, AB must equal 0. Since B = 1, A must = 0.

8. **02-03 C3 Digital Electronics**

The circuit translates as follows: $\overline{A+(\overline{AB}\oplus C)}$ 6

A	B	C	AB	$\overline{AB}$	$\oplus$ C	+ A	NOT
0	0	0	0	1	1	1	0
0	0	1	0	1	0	0	1
0	1	0	0	1	1	1	0
0	1	1	0	1	0	0	1
1	0	0	0	1	1	1	0
1	0	1	0	1	0	1	0
1	1	0	1	0	0	1	0
1	1	1	1	0	1	1	0

There are 6 ordered triples that make it FALSE.

9. **02-03 C4 Digital Electronics**

The three inputs going into the box are: (0,0,1), (0,1,0), (0,1,1)
$\overline{A}+AB$, $\overline{AB}$ and $B+C$

A	B	C	AB	~A	$\overline{A}+AB$	$\overline{AB}$	$B+C$
0	0	0	0	1	1	1	0
0	0	1	0	1	1	1	1
0	1	0	0	1	1	1	1
0	1	1	0	1	1	1	1
1	0	0	0	0	0	1	0
1	0	1	0	0	0	1	1
1	1	0	1	0	1	0	1
1	1	1	1	0	1	0	1

Only (0,0,1), (0,1,0), and (0,1,1) produce all 1's.

10. 03-04 C3 Digital Electronics

The circuit translates to: $\sim(\overline{A+B} \oplus \overline{BC})C$ 31

A	B	C	$A+B$	$\overline{A+B}$	$\overline{BC}$	$\oplus$	$\overline{\overline{A+B} \oplus \overline{BC}}$	$AND\ C$
0	0	0	0	1	1	0	1	0
0	0	1	0	1	1	0	1	1
0	1	0	1	0	1	1	0	0
0	1	1	1	0	0	0	1	1
1	0	0	1	0	1	1	0	0
1	0	1	1	0	1	1	0	0
1	1	0	1	0	1	1	0	0
1	1	1	1	0	0	0	1	1

11. 03-04 C3 Digital Electronics

$\overline{A+B+\overline{BC}} = 1$ so $A+B+\overline{BC} = 0$ implies that A = 0, B = 0 and $\overline{BC} = 0$. 0
This implies that $BC = 1$. That implies B = 1 and C = 1. This causes a contradiction since B must equal 0 and B must equal 1. Therefore there are no triples that make the circuit TRUE.

12. 03-04 C4 Digital Electronics

The circuit translates to $\overline{(\overline{A} \oplus \overline{(A+B)+B\overline{C}})\overline{C}}$. Only (0,1,0) makes it FALSE. (0,1,0)

A	B	C	$\overline{A}$	$A+B$	$B\overline{C}$	+	NOT	$\overline{A} \oplus$	$*\overline{C}$	NOT
0	0	1	1	0	0	0	1	0	0	1
0	0	0	1	0	0	0	1	0	0	1
0	1	1	1	1	0	1	0	1	0	1
0	1	0	1	1	1	1	0	1	1	0
1	0	1	0	1	0	1	0	0	0	1
1	0	0	0	1	0	1	0	0	0	1
1	1	1	0	1	0	1	0	0	0	1
1	1	0	0	1	1	1	0	0	0	1

13. 04-05 C3 Digital Electronics

The circuit translates to: $\overline{A}B \oplus (B+C)$. There are 4 that make it TRUE. 4

A	B	C	$\overline{A}B$	$B+C$	$\oplus$
0	0	0	0	0	0
0	0	1	0	1	1
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	0

1	0	1	0	1	1
1	1	0	0	1	1
1	1	1	0	1	1

14. 04-05 C3 Digital Electronics

The circuit translates directly to: $\overline{\overline{A} + \overline{A \oplus B} + \overline{A \oplus B}\overline{B\overline{C}} + C}$ As shown
Because of the order of operations no parentheses are needed.
An AND (*) operator may be used before the last term.

15. 04-05 C4 Digital Electronics

The circuit translates to: $(\overline{A} \oplus (\overline{A + B} * \overline{BC})) + C$. (0,0,0), (1,0,0),
Three entries make it FALSE. (1,1,0)

A	B	C	$\overline{A}$	$\overline{A+B}$	$\overline{BC}$	*	$\oplus$	$+C$
0	0	0	1	1	1	1	0	0
0	0	1	1	1	1	1	0	1
0	1	0	1	0	1	0	1	1
0	1	1	1	0	0	0	1	1
1	0	0	0	0	1	0	0	0
1	0	1	0	0	1	0	0	1
1	1	0	0	0	1	0	0	0
1	1	1	0	0	0	0	0	1

16. 05-06 C3 Digital Electronics

Sinplifying to: $\overline{\overline{A \oplus B} * (B \oplus C) + \overline{CD}}$ and using a table gives 3 TRUE results. 3

A	B	C	D	A⊕B	NOT	B⊕C	AND	NOT (CD)	+	NOT
0	0	0	0	0	1	0	0	1	1	0
0	0	0	1	0	1	0	0	1	1	0
0	0	1	0	0	1	1	1	1	1	0
0	0	1	1	0	1	1	1	0	1	0
0	1	0	0	1	0	1	0	1	1	0
0	1	0	1	1	0	1	0	1	1	0
0	1	1	0	1	0	0	0	1	1	0
0	1	1	1	1	0	0	0	0	0	1
1	0	0	0	1	0	0	0	1	1	0
1	0	0	1	1	0	0	0	1	1	0
1	0	1	0	1	0	1	0	1	1	0
1	0	1	1	1	0	1	0	0	0	1

1	1	0	0	0	1	1	1	1	1	0
1	1	0	1	0	1	1	1	1	1	0
1	1	1	0	0	1	0	0	1	1	0
1	1	1	1	0	1	0	0	0	0	1

17. 05-06 C3 Digital Electronics

The circuit translates to: $\overline{\overline{\overline{A}+AB}(B+C)C}$ As shown

18. 05-06 C4 Digital Electronics

The circuit translates to $((\overline{AB})(B+\overline{C+D}))+\overline{C+D}=1\Rightarrow$ 7
$\overline{A}B+\overline{A}\,\overline{C}\,\overline{D}+\overline{B}\,\overline{C}\,\overline{D}+\overline{C}\,\overline{D}=1\Rightarrow \overline{A}B+\overline{C}\,\overline{D}=1$. If A=0 and B=1 then C=* and D=*. If C=0 and D=0 then A=* and B=*. The ordered quadruplets are (0,1,*,*) and (*,*,0,0). Each produces 4 results with one overlap (0,1,0,0).

19. 06-07 C3 Digital Electronics

The circuit translates to $(\overline{A}+(A\oplus B))(\overline{B}+C)=1$. 5
Both terms of the AND must be TRUE**.**
If A=1 then B=0 and C=* $\Rightarrow$ (1,0,*)
If A=0, and B=1, then C=1, but if A=0 and B=0 then C=* $\Rightarrow$
(0,1,1) and (0,0,*). Therefore there are 5 solutions.

20. 06-07 C3 Digital Electronics

The circuit translates to $\overline{\overline{A}+AB}+(B+\overline{C})=\overline{(\overline{A}+AB)}(\overline{B+\overline{C}})=$ $\overline{A}\,\overline{B}C$
$(\overline{A}+AB)(\overline{B}C)=\overline{A}\,\overline{B}C+AB\overline{B}C=\overline{A}\,\overline{B}C$

21. 06-07 C4 Digital Electronics

$\overline{\overline{A + \overline{\overline{A}B}}((B + C) \oplus \overline{C})} = 0$ (0,1,1)

$\overline{(A + \overline{\overline{A}B})}((B + C) \oplus \overline{C}) = 1$ Both terms must be TRUE

$A + \overline{\overline{A}B} = 0$ AND $((B + C) \oplus \overline{C}) = 1$

$A = 0$ AND $\overline{\overline{A}B} = 0$ $\quad (1 + C) \oplus \overline{C} = 1$

$\overline{A}B = 1$ $\quad 1 \oplus \overline{C} = 1$

$B = 1$ $\quad \overline{C} = 0 \Rightarrow C = 1$

22. 07-08 C3 Digital Electronics

The circuit translates to $\overline{A(\overline{A}B \oplus (B + C))} + \overline{C}$. Six of them are TRUE. 6

A	B	C	$\overline{A}B$	$B + C$	$\overline{A}B \oplus (B + C)$	$\overline{A(\overline{A}B \oplus (B + C))}$	$+\overline{C}$
0	0	0	0	0	0	1	1
0	0	1	0	1	1	1	1
0	1	0	1	1	0	1	1
0	1	1	1	1	0	1	1
1	0	0	0	0	0	1	1
1	0	1	0	1	1	0	0
1	1	0	0	1	1	0	1
1	1	1	0	1	1	0	0

23. 07-08 C3 Digital Electronics

The circuit translates to $\overline{AB + (B + C)}$ (0,0,0), (1,0,0)

$= \overline{AB}(\overline{B + C}) = (\overline{A} + \overline{B})(\overline{B}\overline{C}) = \overline{A}\overline{B}\overline{C} + \overline{B}\overline{C} = \overline{B}\overline{C}(\overline{A} + 1) = \overline{B}\overline{C}$

$\overline{B}\overline{C} = 1 \rightarrow \overline{B} = 1$ & $\overline{C} = 1 \rightarrow B = 0$ & $C = 0$ & $A = *$

24. 07-08 C4 Digital Electronics

The circuit translates to: $(\overline{A} + (AB \oplus \overline{B + C}))CD$ (0,1,1,1), (0,0,1,1)

To be TRUE, $C = 1$ and $D = 1$ and $(\overline{A} + (AB \oplus \overline{B + C})) = 1$

If $C = 1$, then $(\overline{A} + (AB \oplus \overline{B + 1})) = (\overline{A} + (AB \oplus \overline{1}))$

$= (\overline{A} + (AB \oplus 0))$

Now if $A = 1$, then $\overline{1} + (1B \oplus 0) = 0 + (B \oplus 0)$

This is TRUE if $B = 1$ Hence (1, 1, 1, 1).

Now if $A = 0$, then $\overline{0} + (0B \oplus 0) = 1 + (0 \oplus 0) = 1$

This is TRUE for $B = *$ Hence (0, *, 1, 1).

25. 08-09 C3 Digital Electronics

The circuit translates to $(\overline{A + \overline{A}B})((B + C)\overline{C})$ 0 or FALSE

$$\overline{(A+\overline{A}B)}((B+C)\overline{C}) = (\overline{A}\,\overline{\overline{A}B})(B\overline{C}+C\overline{C}) = \overline{A}(\overline{\overline{A}}+\overline{B})(B\overline{C})$$
$$= (A\overline{A}+\overline{A}\,\overline{B})(B\overline{C}) = \overline{A}\,B\,\overline{B}\,\overline{C} = 0$$

26. 08-09 C3 Digital Electronics

The circuit translates to $\overline{\overline{A}(AB+\overline{C})}+(\overline{C+D})$ (0,1,0,1), (0,0,0,1)

$$\overline{\overline{A}(AB+\overline{C})}+(\overline{C+D}) = \overline{A\overline{A}B+\overline{A}\,\overline{\overline{C}}}+\overline{C}\,\overline{D}$$
$$= \overline{\overline{A}\,\overline{C}}+\overline{C}\,\overline{D} = A+C+\overline{C}\,\overline{D}$$

If $A+C+\overline{C}\,\overline{D}=0$, then $A=0\wedge C=0\wedge\overline{C}\,\overline{D}=0$.

$\overline{0}\overline{D}=1\overline{D}=0\Rightarrow\overline{D}=0\Rightarrow D=1$ Also $B=*$

27. 08-09 C4 Digital Electronics

The circuit translates to: $AB\oplus\overline{B\overline{C}}$ Choice B B, C, D

$AB\oplus\overline{B\overline{C}} = AB\overline{\overline{B\overline{C}}}+\overline{AB}\,\overline{B\overline{C}}$

$= ABB\overline{C}+(\overline{A}+\overline{B})(\overline{B}+C) = AB\overline{C}+\overline{A}\,\overline{B}+\overline{A}C+\overline{B}+\overline{B}C$

$= AB\overline{C}+\overline{B}+\overline{A}C$ Choice C

Choice D simplifies to Choice B using one of the Boolean identities.

28. 09-10 C3 Digital Electronics

The circuit translates to $\overline{((\overline{\overline{A}+B})+(BC))C}$ 5

$$\overline{((\overline{\overline{A}+B})+(BC))C}=1 \Rightarrow \overline{(\overline{A}+B+BC)C}=0$$
$$(A\overline{B}+BC)C=0$$
$$A\overline{B}C+BC=0$$
$$if\ C=0\ then\ A=*\wedge B=*$$
$$if\ C=1\ then\ A\overline{B}+B=0$$
$$\Rightarrow \mathrm{B}=0\wedge\mathrm{A}=0$$

That gives you 5 ordered triples: (1,1,0),(1,0,0),(0,0,0),(0,1,0),(0,0,1).

29. 09-10 C3 Digital Electronics

The circuit translates to $((\overline{AB})(\overline{B+C}))\quad(C\oplus D)\quad D$ 4

A	B	C	D	$\overline{AB}$	$\overline{B+C}$	$(\overline{AB})(\overline{B+C})$	$C\oplus D$	
0	0	0	1	1	1	1	1	1
0	1	0	1	1	0	0	1	1
1	0	0	1	1	1	1	1	1
1	1	0	1	0	0	0	1	1

Space did not allow the whole truth table. So here are the rows that made the expression TRUE. There are 4 of them.

30. 09-10 C4 Digital Electronics

The circuit translates to: $\overline{\overline{A+B}(\overline{B}C \oplus (C+D))}$ 1

To be FALSE, $\overline{A+B}(\overline{B}C \oplus (C+D)) = 1$

$\Rightarrow \quad \overline{A+B} = 1 \wedge \overline{B}C \oplus (C+D) = 1$

$\overline{A+B} = 1 \Rightarrow A+B = 0 \Rightarrow A = 0 \wedge B = 0$

So $(1C \oplus (C+D) = 1$

Now if $C = 1$, then $1+D = 0$ which is impossible

Now if $C = 0$, then $D = 1$

$\therefore$ 1 ordered quadruple makes the expression FALSE

31. 10-11 C3 Digital Electronics

The circuit translates to $\overline{A}(A \oplus B) + \overline{\overline{BC} \oplus C}$. 4

There are 4 that make it TRUE.

A	B	C	$\overline{A}$	$A \oplus B$	$\overline{A}(A \oplus B)$	$\overline{BC}$	$\overline{\overline{BC} \oplus C}$	$\overline{A}(A \oplus B) + \overline{\overline{BC} \oplus C}$
0	0	0	1	0	0	1	0	0
0	0	1	1	0	0	1	1	1
0	1	0	1	1	1	1	0	1
0	1	1	1	1	1	0	0	1
1	0	0	0	1	0	1	0	0
1	0	1	0	1	0	1	1	1
1	1	0	0	0	0	1	0	0
1	1	1	0	0	0	0	0	0

32. 10-11 C3 Digital Electronics

This circuit translates to $\overline{(AB+\overline{B}) + \overline{B+C}}$ $\overline{A}B$

$\overline{(AB+\overline{B}) + \overline{B+C}} = \overline{AB+\overline{B}}\,\overline{\overline{B+C}} = (\overline{AB})\overline{\overline{B}}(B+C) = (\overline{A}+\overline{B})B(B+C)$

$= (\overline{A}B + \overline{B}B)(B+C) = \overline{A}B(B+C) = \overline{A}B + \overline{A}BC = \overline{A}B(1+C) = \overline{A}B$

33. 10-11 C4 Digital Electronics

The circuit translates to: $\overline{(A+(A+B)C)(CD+D)}$ 5

$\overline{(A+(A+B)C)(CD+D)} = \overline{(AC+BC+A)(CD+D)}$

$= \overline{(A(C+1)+BC)(D(C+1)} = \overline{(A+BC)D}$

$= \overline{AD+BCD}$

The Truth Table for the above produces 5 FALSE results.

34. 11-12 C4 Digital Electronics

$\overline{A(A\oplus B)}+(B+C)C$ $= \overline{A}+\overline{A\oplus B}+BC+C$ (1,0,0)

$= \overline{A}+\overline{A}\oplus B+BC+C$

$= \overline{A}+\overline{A}\,\overline{B}+\overline{\overline{A}B}+BC+C$

$= \overline{A}(1+\overline{B})+AB+C(B+1)$

$= \overline{A}+AB+C$

$\overline{A}+AB+C=0 \Rightarrow \overline{A}=0 \wedge AB=0 \wedge C=0 \Rightarrow$

$A=1 \wedge B=0 \wedge C=0$

35. 11-12 C4 Digital Electronics

$((\overline{A+B})(\overline{BC}))(\overline{C+D})$ $= (\overline{A+B}+\overline{BC})(\overline{C}\,\overline{D}) = (\overline{A}\,\overline{B}+\overline{B}+\overline{C})(\overline{C}\,\overline{D})$ $\overline{C}\,\overline{D}$

$= \overline{A}\,\overline{B}\,\overline{C}\,\overline{D}+\overline{B}\,\overline{C}\,\overline{D}+\overline{C}\,\overline{D} = \overline{C}\,\overline{D}$

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36. 12-13 C4 Digital Electronics

The circuit translates to: $\overline{(AB)(B+C)+\overline{\overline{C}}}$ (0,0,1), (0,1,1)

$\overline{(AB)(B+C)+\overline{\overline{C}}} = \overline{(AB)(B+C)}C = (\overline{AB}+\overline{B+C})C$ (1,0,1)

$= ((\overline{A}+\overline{B})+\overline{B}\,\overline{C})C = \overline{A}C+\overline{B}C+B\overline{C}C = (\overline{A}+\overline{B})C$

So $(\overline{A}+\overline{B})C=1 \Rightarrow \overline{AB}=1 \wedge C=1 \Rightarrow AB=0 \wedge C=1$

Therefore (0, 0, 1), (0, 1, 1), (1, 0, 1) make the circuit true.

37. 12-13 C4 Digital Electronics

Circuit translates $(A+\overline{(A+B)(BC)}\oplus((\overline{C+D})\overline{D})$ and simplifies to: 6

$(A+\overline{B}+\overline{C})\oplus\overline{C}\,\overline{D}$. Of the 16 possible, 6 make if FALSE.

A	B	C	D	$\overline{B}$	$\overline{C}$	$\overline{D}$	$A+\overline{B}+\overline{C}$	$\overline{C}\,\overline{D}$	$\oplus$

0	0	0	0	1	1	1	1	1	0
0	0	0	1	1	1	0	1	0	1
0	0	1	0	1	0	1	1	0	1
0	0	1	1	1	0	0	1	0	1
0	1	0	0	0	1	1	1	1	0
0	1	0	1	0	1	0	1	0	1
0	1	1	0	0	0	1	0	0	0
0	1	1	1	0	0	0	0	0	0
1	0	0	0	1	1	1	1	1	0
1	0	0	1	1	1	0	1	0	1
1	0	1	0	1	0	1	1	0	1
1	0	1	1	1	0	0	1	0	1
1	1	0	0	0	1	1	1	1	0
1	1	0	1	0	1	0	1	0	1
1	1	1	0	0	0	1	1	0	1
1	1	1	1	0	0	0	1	0	1

38. 13-14 C4 Digital Electronics

The circuit translates to: $(\overline{AB})(\overline{B+C})\oplus C$ (*.*.1), (*,0,0)

$(\overline{AB})(\overline{B+C})\oplus C=(\overline{A}+\overline{B})(\overline{B}\,\overline{C})\oplus C=(\overline{A}\,\overline{B}\,\overline{C}+\overline{B}\,\overline{B}\,\overline{C})\oplus C=\overline{B}\,\overline{C}\oplus C$ (1,1,1),(1,0,1),

So $\overline{B}\,\overline{C}\oplus C=1$. $C=1\rightarrow\overline{B}\,\overline{C}=0\rightarrow B=*,A=*,C=1$ So (*, *, 1) (0,1,1),(0,0,1),

$C=0\rightarrow\overline{B}\,\overline{C}=1\rightarrow B=0,A=*,C=0$ So (*, 0, 0) (1,0,0),(0,0,0)

39. 13-14 C4 Digital Electronics

Circuit translates $\overline{(\overline{A}+AB)+\overline{((\overline{B+C})(CD))}D}$ $A\overline{B}\,\overline{C}D$

$= (\overline{\overline{A}+AB})(\overline{\overline{((\overline{B+C})(CD))}D}=(\overline{\overline{A}}\,\overline{AB})((\overline{\overline{B+C}})(\overline{CD})D$

$= A(\overline{A}+\overline{B})((\overline{B+C})+\overline{CD})D=(AD\overline{A}+AD\overline{B})(\overline{B}\,\overline{C}+\overline{C}+\overline{D})$

$= AD\overline{B}\,\overline{B}\,\overline{C}+AD\overline{B}\,\overline{C}+AD\overline{B}\,\overline{D}=A\overline{B}\,\overline{C}D$

40. 14-15 C4 Digital Electronics

$\overline{(\overline{A}B)(\overline{B}\,\overline{C})}+(\overline{B}\,\overline{C}B\oplus BC)=(\overline{\overline{A}B}+\overline{\overline{B}\,\overline{C}})+(((\overline{B}+\overline{C})B)\oplus BC)$ 8

$=((\overline{\overline{A}}+\overline{B})+BC)+((\overline{B}B+\overline{C}B)\oplus BC)=(A+\overline{B}+BC)+(\overline{B}B\oplus BB)$

$=(A+\overline{B}+BC)+(0\oplus B)=(A+\overline{B}+BC)+(0\overline{B}+\overline{0}B)$

$=(A+B+BC)+(0+1)$

$=(A+\overline{B}+BC)+1=1$

Therefore all triples will make it TRUE.

41. 14-15 C4 Digital Electronics

$\overline{(\overline{A} \oplus \overline{\overline{A}B}) + (\overline{B\overline{C} + C})} = (\overline{\overline{A} \oplus \overline{\overline{A}B}})(\overline{\overline{B\overline{C} + C}})$ (0,0,1)

$= (\overline{\overline{A}\overline{\overline{A}B} + \overline{\overline{A}}\overline{\overline{A}B}})(B\overline{C} + C) = (\overline{A}(\overline{\overline{A}} + \overline{B}))(B\overline{C} + C) = (\overline{A}A + \overline{A}\overline{B})(B\overline{C} + C)$

$= \overline{A}\overline{B}B\overline{C} + \overline{A}\overline{B}C = \overline{A}\overline{B}C$

Therefore (0,0,1) makes it TRUE.

42. 15-16 C4 Digital Electronics

The circuit translates to: $\overline{\overline{A}\overline{\overline{A}B} + (B + \overline{C})}C$ $A\overline{B}C$

$\overline{\overline{A}\overline{\overline{A}B} + (B + \overline{C})}C = \overline{\overline{A}(\overline{\overline{A}} + \overline{B}) + (B + \overline{C})}C = \overline{\overline{A}A + \overline{A}\overline{B} + B + \overline{C}}C =$

$\overline{\overline{A}\overline{B} + B + \overline{C}}C = \overline{\overline{A}}\overline{\overline{B}}\overline{B}CC = (\overline{\overline{A}} + \overline{\overline{B}})\overline{B}C = A\overline{B}C + B\overline{B}C = A\overline{B}C$

43. 15-16 C4 Digital Electronics

The circuit translates to: $\overline{\overline{(\overline{A} + BC) + (\overline{CD} + D)}}$ 0

$\overline{(\overline{A} + BC) + (\overline{CD} + D)} = \overline{\overline{(\overline{A} + BC)}}\overline{(\overline{CD} + D)} = (\overline{A} + BC)(\overline{\overline{CD}}\overline{D})$

$= (\overline{A} + BC)CD\overline{D} = 0$